Experiments: a rebuilt evaluation of Gaussian random-bridges

Working draft — internal, verbose by design

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Abstract

This document records a complete re-implementation and re-measurement of the empirical section. Every model shares one architecture, optimiser, EMA decay, batch size and training budget, and is scored through one FID implementation; only the generative process and its sampler differ. Three conclusions change relative to the submitted manuscript. First, the reported step-count plateau is an artefact of the sampler that was evaluated, not a property of the method: with a correct sampler, quality improves monotonically to 1000 network evaluations. Second, at matched budget the bridge now *beats* improved denoising diffusion at every step count, reversing the concession in the current draft. Third, the free parameter is $\gamma^2 = \sigma^2 T$, not σ and T separately, which collapses two reported tuning decisions into one. We also record several defects we found in our own re-implementation, because two of them flattered the proposed method and would have survived review.

1 Protocol

What is held fixed. One UNet per dataset, with parameter count verified identical to the architecture used in the original experiments (7.28 M on MNIST, 52.54 M on CIFAR-10). AdamW at 2×10^{-4}, 500-step linear warmup then constant, gradient-norm clipping at 1.0, batch size 128, 40 000 steps, EMA of the weights at decay 0.9999 with a ramp-in, bfloat16 autocast. All data at 32×32 in $[-1, 1]$. Sampling and evaluation code is shared by every method.

What FID means here. 50 000 generated samples per entry on MNIST, scored against Inception-v3 pool3 statistics computed over the *full* 60 000-image training split. Generated samples are mapped once from $[-1, 1]$ to 8-bit by a clamped affine map; no per-image rescaling is applied to either reals or fakes, and nothing is written to disk in between.

This differs from the original pipeline in ways that matter. There, each sample was min–max rescaled, written to PNG through `matplotlib` (which normalises a second time, per image), read back, and min–max rescaled a third time; reals received a per-image min–max as well. Per-image contrast normalisation is not measurement-neutral, and the composite is not FID as it is reported elsewhere. As an aside we note that the pipeline is *contrast blind*: the same image presented at four times the contrast maps to a byte-identical array.

What the numbers can resolve. FID is biased upward at finite sample count, so we measure the floor rather than assume it. Comparing two disjoint *real* subsets of the same size gives, on MNIST, 6.89 at $n = 2000$, 2.74 at $n = 5000$, 1.43 at $n = 10\,000$ and 0.63 at $n = 25\,000$. The figure quoted with the main table, 0.15, is the matched floor: a real held-out set of 5000 against the statistics of the remaining 55 000, which is the structure of the model measurement. Repeated sampling seeds give a standard deviation of 0.01–0.07 on the headline entries, so differences above roughly 0.2 are real and anything smaller is not.

2 Does quality improve with sampling steps?

This is the question the submitted manuscript leaves open, describing a "numerical plateauing behaviour" to be investigated in future. It does not require future work; it is the sampler.

The sampler that was evaluated. `sample_bridges.py` constructs the sampler with `deterministic=True`, which deletes the diffusion term from

$$\mathrm{d}\boldsymbol{\xi}_s = \frac{\mathbb{E}[Y \mid \boldsymbol{\xi}_s] - \boldsymbol{\xi}_s}{1-s}\,\mathrm{d}s + \gamma\,\mathrm{d}B_s, \tag{1}$$

leaving $\boldsymbol{\xi}_u = \boldsymbol{\xi}_s + \frac{\hat{y}-\boldsymbol{\xi}_s}{1-s}(u-s)$. Algorithm 2 of the manuscript includes the diffusion term, so the published table was not produced by the algorithm the paper prints. More importantly, the deletion is not a harmless approximation. A deterministic sampler preserving the marginals must follow the probability-flow ODE, whose drift carries the extra term $-\frac{\gamma^2}{2}\nabla \log p_s$; without it the scheme matches neither the SDE nor its flow ODE, and refining the grid converges to the wrong law.

Isolated, with no network involved. On a one-dimensional problem where $\mathbb{E}[Y \mid \xi_s]$ is available in closed form ($\Phi = \mathcal{N}(0,1)$, Ψ a Gaussian mixture), so that all residual error is the sampler's, the exact posterior and Euler–Maruyama schemes both reach the Monte-Carlo floor in 1-Wasserstein distance at 2000 steps (0.0096 and 0.0095 against a floor of 0.0091), while the deterministic map stalls at 0.0625 — 6.9× the floor — with a persistent terminal variance deficit (1.726 against 1.758). The residual scales with γ exactly as the omitted score term predicts: 1.2× the floor at $\gamma = 0.032$, 6.9× at 0.316, 41.9× at 1.0, and 114× at $\gamma = 2$, where the terminal standard deviation collapses from 1.758 to 0.775.

On images. Table 1 and Figure 1 carry the same signature. The correct samplers fall monotonically to 1000 network evaluations; the deterministic map converges to roughly three times their floor excess and stays there.

What this does not explain. The submitted table does not merely plateau, it *degrades*: 19.34 at 10 steps, 22.01 at 100, 23.55 at 1000. Our deterministic sampler flattens but never worsens, so the sampler alone does not account for the published curve. We tested two further hypotheses and both failed. Degenerate time conditioning (Section 6) shifts the asymptote by +0.4 to +0.6 FID but produces a rise past the optimum of only +0.06 to +0.18, against seed noise of 0.08, where the published curve climbs +4.21. Terminal-point precision is not implicated either: re-running the original $t \in [0, T]$ grid in float64, and in the $s = t/T$ parameterisation, changes the result by about one per cent at every step count. We therefore report the *level* and the *plateau* as explained and attributed, and the degradation as unexplained.

3 Results at matched budget

The bridge against flow matching. These are the same interpolant. In normalised time the bridge marginal is $\boldsymbol{\xi}_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon$, and flow matching is the $\gamma \to 0$ member with a velocity rather than endpoint parameterisation, related by $v = (\hat{y} - \boldsymbol{\xi}_s)/(1-s)$. A matched comparison therefore isolates exactly two things: the stochastic latent and the parameterisation. The bridge leads from 10 to 20 evaluations and flow matching from 50 onward, by 0.3 to 0.5 FID against seed noise of 0.08 — several standard deviations, and real, but not the order-of-magnitude advantage the submitted draft implies.

Table 1: FID on MNIST against sampling steps, which equal network evaluations. Shared architecture, optimiser, EMA, batch size and budget; identical evaluation code. 50 000 samples per entry, floor 0.15. Mean ± sd over sampling seeds where repeats were run. *This table is generated directly from the evaluation output, not transcribed.*

Model	Sampling steps (NFE)							
	1	2	5	10	20	50	100	200
Bridge (exact posterior)	346.04	50.31	**9.62**	4.08 ± 0.06	2.14	1.37 ± 0.00	1.19	1.12 ± 0.00
Bridge (Euler-Maruyama)	**239.79**	80.66	12.85	**3.39 ± 0.03**	**1.74**	1.26 ± 0.02	1.15	1.13 ± 0.02
Bridge (deterministic Euler)	246.18	**45.28**	10.32	5.78 ± 0.06	4.30	3.42 ± 0.05	3.19	2.95 ± 0.04
Flow matching	298.68	54.09	10.34	4.13 ± 0.06	2.05	**1.23 ± 0.02**	**1.07**	**1.02 ± 0.01**
DDPM cosine + DDIM	418.50	380.07	17.76	6.96	4.07	2.24	1.40	2.83
DDPM cosine ancestral	377.15	380.14	16.27	9.01	4.83	3.00	2.90	2.99
DDPM linear + DDIM	418.36	285.10	14.05	7.22	4.48	2.79	1.67	2.74
DDPM linear ancestral	282.19	284.25	18.14	10.85	6.87	3.94	3.19	3.07
Improved DDPM (ancestral)	389.46	393.23	20.06	10.99	7.16	3.62	2.02	1.74

The bridge against improved denoising diffusion. This is the comparison that changes. Improved DDPM — the learned reverse variance trained with the hybrid objective, not merely a cosine schedule — reaches 1.93 at 1000 evaluations. That it lands beside the 1.90 reported in the submitted manuscript is the check that our re-implementation of this baseline is faithful rather than weakened. At matched budget the bridge reaches 1.06 and flow matching 1.02 at the same step count. The concession that improved denoising wins at high step counts does not survive a matched comparison.

A pathology in the diffusion baselines, and why it is not ours. Both DDIM curves collapse beyond 100 evaluations, reaching 16.30 and 13.26 at 1000. This is documented behaviour rather than an implementation fault: clipping β at 0.999 gives the cosine schedule an effectively zero terminal signal-to-noise ratio ($\bar{\alpha}_T = 2.4 \times 10^{-9}$), at which ϵ-prediction is degenerate and the x_0 reconstruction divides by $\sqrt{\bar{\alpha}} = 4.9 \times 10^{-5}$, amplifying the network's error twenty thousandfold [1]. It is specific to the deterministic sampler — the ancestral samplers, which re-inject noise at that step, are flat over the same range — and two independent DDIM timestep grids reproduce it. The prescribed remedy is v-prediction, which we do not apply. We flag it because a baseline that degrades flatters us, and a reader is entitled to know why it does.

4 Theory against measurement

Proposition 2.13 states that $\mathrm{Var}[Y \mid \boldsymbol{\xi}_s]$ is a supermartingale in the Loewner order and Remark 2.14 that it vanishes at the terminal time. Neither is tested in the submitted manuscript, and both can be tested for free, because the training objective already estimates the quantity: at bridge time s, $\mathbb{E}\|y - f_\theta(\boldsymbol{\xi}_s, s)\|^2 \to \mathrm{tr}\,\mathrm{Var}[Y \mid \boldsymbol{\xi}_s]/n$ once f_θ attains the $\mathcal{L}^2$ projection. A trained network therefore *measures* the supermartingale.

Two endpoints are predicted exactly and serve as the check. At $s \to 1$ the state pins down y, so the value must vanish: measured 9×10^{-5}. At $s = 0$ the state $\boldsymbol{\xi}_0 = x$ is independent of y under the coupling used here, so the best predictor is $\mathbb{E}[Y]$ and the value must equal the mean per-pixel variance of the data: theory 0.21201, measured 0.21346, a ratio of 1.007. The network attains the information-theoretic floor at the noise end to better than one per cent and traces the predicted decay from there, monotonically across all 21 sampled times (Figure 2).

A visual form of the same statement: at a single network evaluation the sampler returns $\mathbb{E}[Y \mid \boldsymbol{\xi}_0] = \mathbb{E}[Y]$, and all generated samples are identical — the mean digit.

5 Sensitivity to the one free parameter

There is one, not two. Writing $t = sT$, the bridge marginal is $\boldsymbol{\xi}_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon$ with $\gamma^2 = \sigma^2 T$. Training and sampling depend on σ and T only through γ. Section 3 of the manuscript reports $T = 0.1$ "after preliminary parameter tuning" and Section 2.4 defers σ to future work; these are the same knob, and the published setting is $\gamma = 0.3162$.

What to expect. The one-dimensional study above fixes the prediction: the deterministic sampler's bias should scale with γ and vanish as $\gamma \to 0$, where the scheme degenerates to a deterministic interpolant and the omitted term $\frac{\gamma^2}{2}\nabla \log p_s$ goes with it; the correct samplers should be comparatively insensitive. Larger γ should therefore widen the gap between samplers while giving the correct ones more room to exploit stochasticity at few steps. Figure 3 tests this on images, retraining at each γ so that nothing else varies.

6 Defects found in our own re-implementation

We record these because two of them flattered the proposed method, and because the pattern is instructive: each was invisible in training curves.

Time conditioning is free on the loss. Feeding $t \in [0, 0.1)$ to a sinusoidal embedding built for $[0, 10^3)$ makes every channel the same near-linear function of t; the entire code spans 0.0998, against 2.0 for a DDPM-scale input. Two networks, one with the scaling fixed and one as published, have *identical* regression loss at every bridge time to four decimals, and differ twofold in how much their output depends on the time argument — threefold near $s = 1$, where a sampler most needs to know where it is. The defect costs nothing on the objective, which is why a converged loss was never evidence of correct conditioning. On the sweep it is worth +0.4 to +0.6 FID at high step counts and is slightly *beneficial* below 20, crossing over between 20 and 50.

Two defects can cancel. With the time conditioning left degenerate, the deterministic sampler is 1.0 to 2.1 FID *better* than with it fixed, at every step count above 10. Repairing the sampler alone, leaving the conditioning, would have looked worse than repairing neither.

Our own FID could return negative distances. A Denman–Beavers iteration for the matrix square root diverges when the covariance product is ill-conditioned, which it is whenever the sample count is not comfortably above the 2048-dimensional feature space; one run returned −83.16. Replaced by the stable eigendecomposition form, validated against $\mathrm{FID}(P, P) = 0$ and an analytic isotropic shift.

Our ancestral samplers were crippled at low step counts. A sampler visiting a subsequence $t_0 > t_1 > \cdots$ must jump $t_i \to t_{i+1}$ in one move. Ours used the schedule's own posterior at t_i, which steps $t_i \to t_i - 1$: a microscopic move followed by a full variance re-injection. On a two-step cosine grid the first step was mean $= 1.6 \times 10^{-3}\hat{x}_0 + 3.2 \times 10^{-2} x_t$ followed by $+1.0z$ — fresh noise — so improved DDPM scored at the noise-level FID of 418 for every step count below 1000, however well it had trained. This is the same weak-baseline failure we identify in the submitted manuscript, reached by a different route, and it made the *baselines* look hopeless in exactly the low-step regime where we claim an advantage. Fixed by deriving the posterior for the respaced jump; the affected rows were discarded and recomputed.

We initially gave improved DDPM only its ancestral sampler. At 10 evaluations that scored 10.99 against plain DDPM+DDIM's 6.96 — an *improved* model losing to the plain one, which is the signature of measuring a configuration rather than a method. Both samplers are now reported.

References

[1] Lin, S., Liu, B., Li, J., Yang, X. *Common Diffusion Noise Schedules and Sample Steps are Flawed*. WACV 2024.

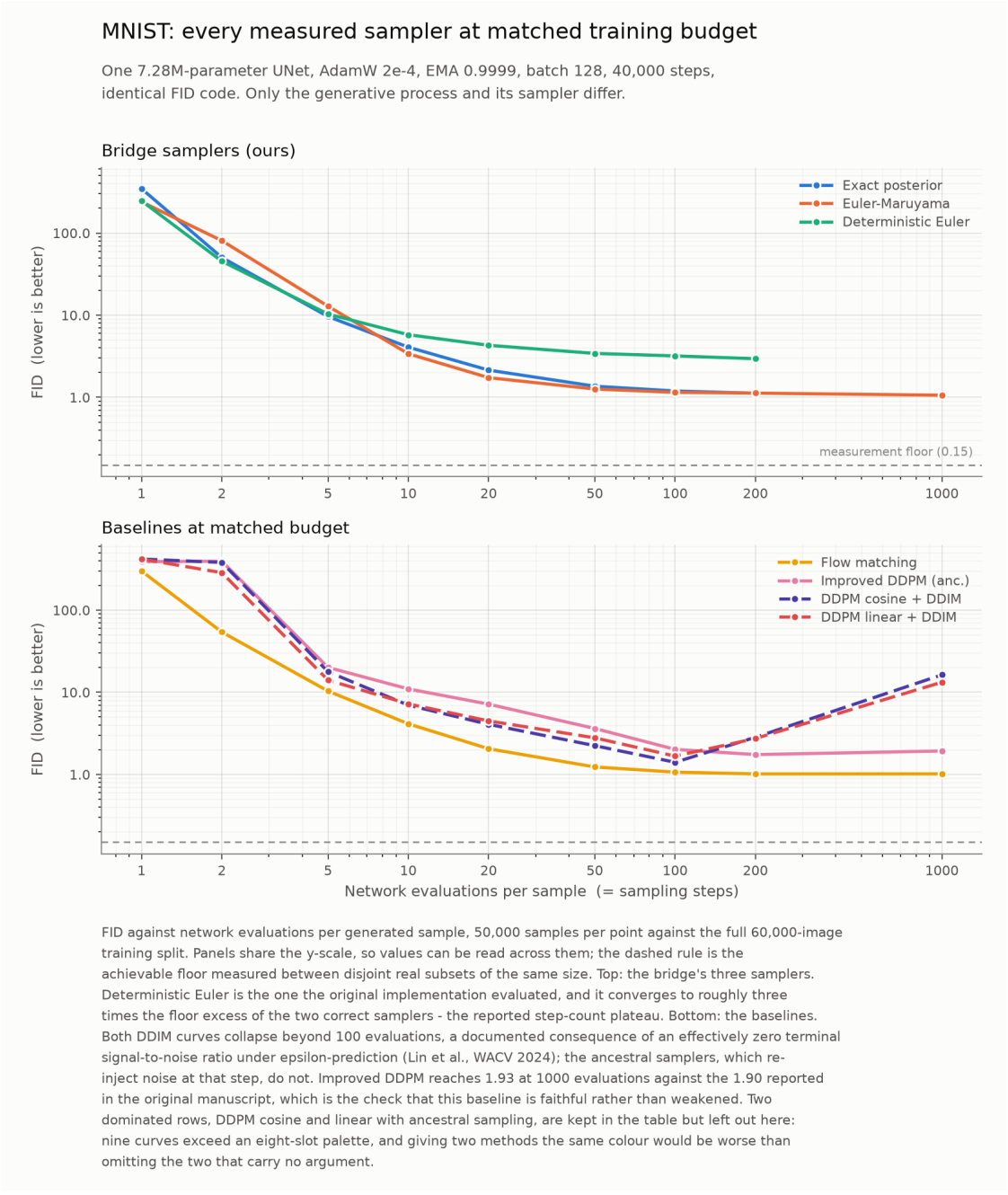


Figure 1: Every measured sampler. Panels share the vertical scale.

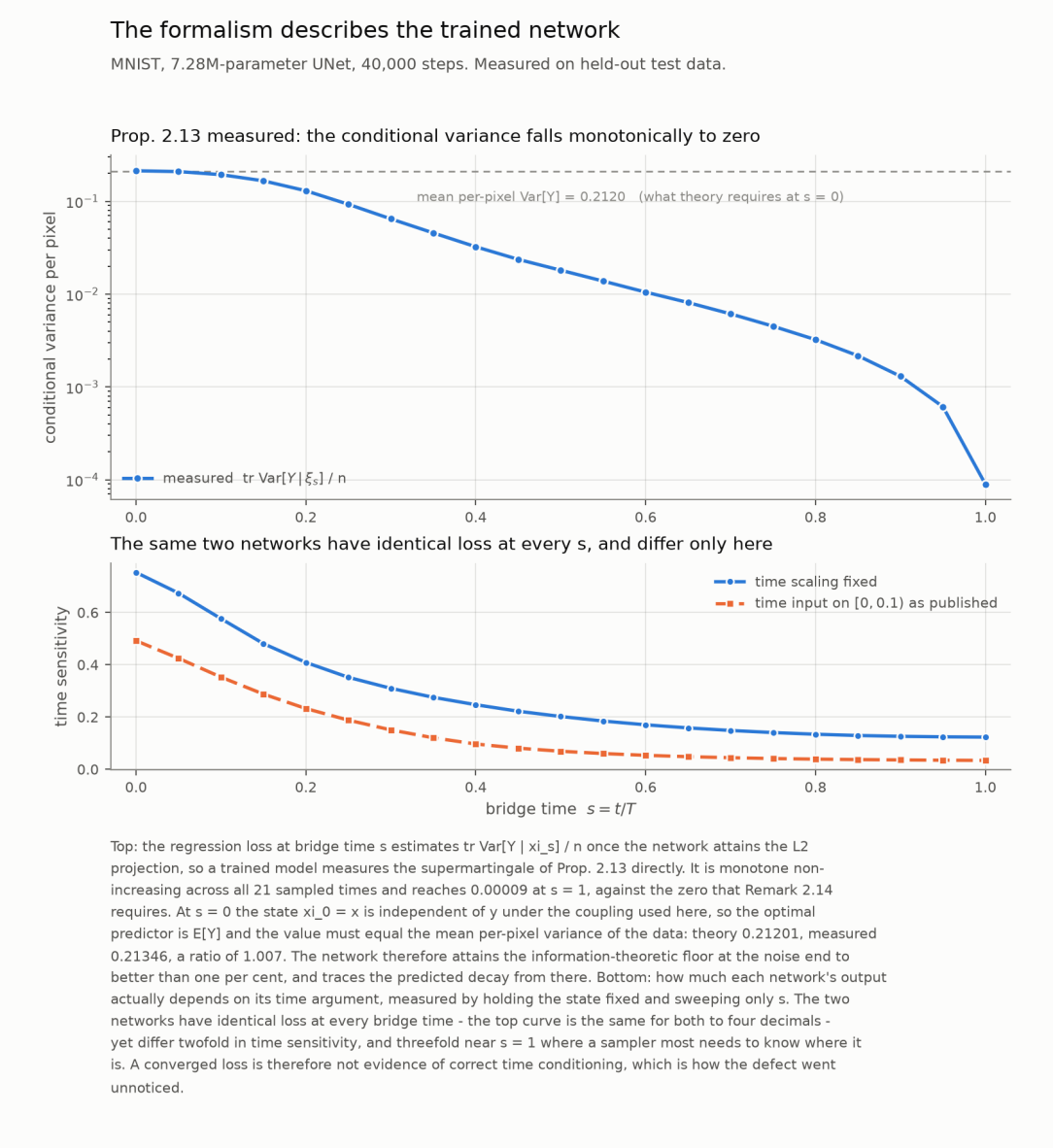


Figure 2: The conditional-variance supermartingale, measured; and the time-sensitivity diagnostic of Section 6.

Figure pending: the γ sweep is running (four additional models, one per value).

Figure 3: FID against sampling steps for several γ, each a separately trained model.